Piecewise-Deterministic Markov Processes

Theory and Methods for Stationary Distributions

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Outline

Motivation and Context

Definition of the PDMP

Forward Kolmogorov Equation

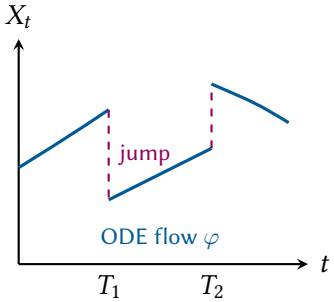
Stability, ergodicity

Appendix: Mellin Transform Analysis of the TCP/IP Stationary Distribution

Piecewise deterministic Markov processes

Davis (1984) observed that continuous-time stochastic models consist of:

1. **Diffusion** (Brownian noise)
2. **Deterministic motion**
3. **Random jumps**



The Gap

Diffusion theory is highly unified (Itô calculus). Non-diffusion models — ingredients (2)+(3) — were a *heterogeneous collection of special cases.*

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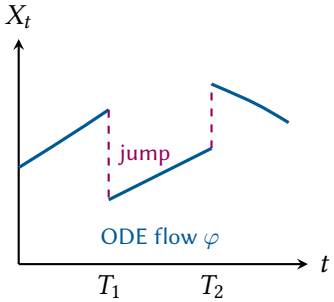


The Gap

Diffusion theory is highly unified (Itô calculus). Non-diffusion models — ingredients (2)+(3) — were a *heterogeneous collection of special cases*.

Goal: Provide a *canonical class* covering virtually all non-diffusion models, with a unified stochastic calculus analogous to diffusion theory.

Motivating Examples

Classical examples (Davis 1984)

- Countable-state Markov processes
- Non-exponential sojourn-time processes
- $M/G/1$ and $GI/G/1$ queues (virtual waiting time)
- Dam / storage / release models

Modern applications

- TCP/IP congestion control
- Reliability & maintenance models
- Biochemical reaction networks
- Risk / ruin theory (insurance)

Key claim (Davis 1984)

The class is *"wide enough to include as special cases virtually all the non-diffusion models of applied probability."*

Local Characteristics

A PDMP on $E \subseteq \mathbb{R}^d$ (open) is determined by three objects:

(1) The flow φ

Unique solution map of $\dot{x} = f(x)$, so

$$\varphi(0, x) = x$$

$$\varphi(t+s, x) = \varphi(t, \varphi(s, x))$$

Assumes f is Lipschitz continuous.

Local Characteristics

A PDMP on $E \subseteq \mathbb{R}^d$ (open) is determined by three objects:

(2) The jump rate function λ

Measurable $\lambda : E \to \mathbb{R}_+$.

Integrated rate:

$$\Lambda(t, x) = \int_0^t \lambda(\varphi(s, x))\, \mathrm{d}s$$

governs the Poisson-like clock.

Local Characteristics

A PDMP on $E \subseteq \mathbb{R}^d$ (open) is determined by three objects:

(3) The jump kernel Q

Probability measure $Q(x, \cdot)$ on E.

Post-jump location

$$Z_1 \sim Q(\varphi(T_1, x_0), \cdot)$$

drawn independently of T_1.

The Integrated Hazard Rate Λ

The function $\lambda : E \to \mathbb{R}_+$ is a **hazard rate**: it measures the instantaneous intensity of jumping *at the current location of the flow.*

Because the process moves deterministically between jumps, the relevant object is the **integrated hazard along the trajectory**:

$$\Lambda(t, x) \;=\; \int_0^t \lambda\big(\varphi(s, x)\big)\, \mathrm{d}s.$$

This is the cumulative intensity seen by a particle starting at x and following the ODE flow for time t.

The first jump time T_1 is then defined as a *non-homogeneous exponential* with this integrated rate:

$$P(T_1 > t \mid X_0 = x) \;=\; e^{-\Lambda(t,\, x)}.$$

Boundary condition

- **Reachable boundary:**
  $\partial^* E = \{z \in \partial E : \varphi(-t, z) \in E \text{ for all } t \in (0, \varepsilon), \text{ some } \varepsilon > 0\}$ — boundary points that the flow can reach from inside E.
- **Boundary exit time:** $t^*(x) = \inf\{t > 0 : \varphi(t, x) \in \partial^* E\}$, with $t^*(x) = \infty$ if the flow never reaches $\partial^* E$.

Davis's Assumption (boundary jumps)

If the flow reaches the boundary in finite time, i.e. $t^*(x) < \infty$, Davis assumes

$$\Lambda(t^*(x),\, x) = \infty.$$

Consequence: $P(T_1 < t^*(x)) = 1$, so the Poisson clock *always fires before the boundary is reached*. Forced boundary jumps are excluded.

Remark: Costa (1990) relaxes this assumption.

Sample Path Construction

Starting from $x_0 \in E$, one inter-jump cycle is constructed as follows.

1. **Draw the alarm level.** Sample $U \sim \mathrm{Uniform}(0,1)$ and set $s^* = -\log U > 0$.
2. **Run the flow and clock jointly.** Integrate $\dot{x} = f(x)$ from x_0 while simultaneously accumulating $\Lambda(t, x_0) = \int_0^t \lambda(\varphi(s, x_0))\,\mathrm{d}s$. Stop at the first time

$$T_1 = \inf\left\{t > 0 : \Lambda(t, x_0) \geq s^*\right\}.$$

   Under Davis's assumption, $T_1 < t^*(x_0)$ with probability 1.
3. **Draw the post-jump location.** Sample $Z_1 \sim Q\left(\varphi(T_1, x_0), \cdot\right)$ independently of T_1.
4. **Restart.** Set $X_{T_1} = Z_1$ and repeat from step 1.

Markov Property and Non-Explosion

Markov property

The semigroup property of φ ensures the process is **strong Markov**: the distribution of the next jump time depends only on the current state X_t, not on the history.

Non-explosion (Davis, Assumption 3.1)

Require $E[N_t] < \infty$ for all $t \geq 0$, equivalently $T_n \to \infty$ a.s., so the process does not accumulate infinitely many jumps in finite time.

Remark

Non-explosion is not automatic — it must be verified for each model.

Example: TCP/IP Congestion Control

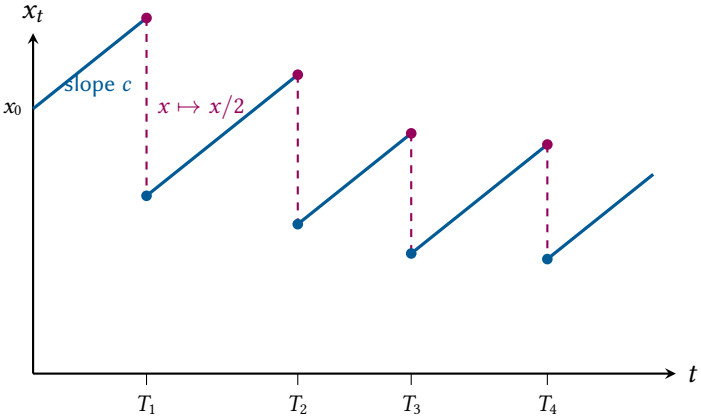
Informal description

A TCP flow increases its sending rate continuously (deterministic motion) and suffers random packet loss events (jumps) that halve the rate — the classical AIMD mechanism.

Local characteristics:

- **State space:** $E = \mathbb{R}_+$, where x is the current sending rate.
- **Flow:** $f(x) = c > 0$ (additive increase), so $\varphi(t, x) = x + ct$.
- **Jump rate:** $\lambda(x) = \alpha x$, $\alpha > 0$ — loss probability grows with the sending rate.
- **Jump kernel:** $Q(x, \cdot) = \delta_{x/2}$ — upon a loss event the rate is halved.

Example: TCP/IP Congestion Control



The Generator: Motivation

We want to characterize how $E[\psi(X_t)]$ evolves. For a smooth test function $\psi : E \to \mathbb{R}$, track $\psi(X_t)$ along a sample path.

Between jumps: $X_t = \varphi(t, x)$ follows the ODE, so by the chain rule:

$$\frac{d}{dt}\psi(\varphi(t,x)) = f(\varphi(t,x)) \cdot \nabla\psi(\varphi(t,x)).$$

The instantaneous rate of change is the **directional derivative of ψ along the vector field f.**

At a jump time T_i: the process jumps from $X_{T_i^-}$ to $Z_i \sim Q(X_{T_i^-}, \cdot)$, so ψ changes by:

$$\psi(Z_i) - \psi(X_{T_i^-}).$$

Taking expectations, the average jump contribution at x is:

$$\lambda(x) \int_E [\psi(y) - \psi(x)]\, Q(x, \mathrm{d}y).$$

The Generator: Domain $\mathcal{D}(\mathcal{A})$

Not every function ψ can be in the domain of $\mathcal{A}$. Davis identifies three conditions.

1. **Absolute continuity along trajectories.** $t \mapsto \psi(\varphi(t,x))$ is absolutely continuous on $[0, t^*(x))$ for each $x \in E$.
   Why: ensures the directional derivative $f(x) \cdot \nabla\psi(x)$ exists Lebesgue-a.e. along each trajectory, so the ODE contribution is well defined.

2. **Boundary condition.** $\psi(x) = \int_E \psi(y)\, Q(x, \mathrm{d}y)$ for all $x \in \partial^* E$.
   Why: relevant when Davis's assumption is relaxed and forced boundary jumps can occur (Costa 1990). Under Davis's assumption this condition is vacuous.

3. **Integrability.** $\mathcal{B}\psi \in L^{1,\mathrm{loc}}(p)$, where $\mathcal{B}\psi(x) = \int_E [\psi(y) - \psi(x)]\, Q(x, \mathrm{d}y)$.
   Why: ensures the stochastic integral against the jump martingale measure is well defined.

The Extended Generator: Davis (1984) Theorem 5.5

Theorem (Davis 1984, Theorem 5.5)

For $\psi \in \mathcal{D}(\mathcal{A})$, the extended generator of the PDMP is:

$$\boxed{\mathcal{A}\psi(x) \;=\; \underbrace{f(x)\cdot\nabla\psi(x)}_{\textit{ODE drift}} \;+\; \underbrace{\lambda(x)\int_E \big[\psi(y)-\psi(x)\big]\,Q(x,\mathrm{d}y)}_{\textit{average jump contribution}}}$$

Martingale connection

Dynkin formula

For $\psi \in \mathcal{D}(\mathcal{A})$, the process

$$C_t^{\psi} \;=\; \psi(X_t) - \psi(X_0) - \int_0^t \mathcal{A}\psi(X_s)\,\mathrm{d}s$$

is a local $\mathcal{F}_t$-martingale. This is the fundamental connection between the generator and the process.

- $\mathcal{A}$ is not a local operator — the integral term couples $\psi(x)$ to all states y reachable by a jump.
- Unlike diffusion, the calculus here is *ordinary* calculus with a stochastic interpretation — no Itô correction terms.

Kolmogorov Forward Equation: Heuristic

Let $p(x,t)$ denote the density of X_t. Track the probability mass in a small region $A \subset E$ over a small time interval $[t, t+dt]$:

- **Loss due to flow.** The ODE carries mass out of A through its boundary at rate $f(x) \cdot n(x)$ where $n(x)$ is the outward normal. By the divergence theorem:

$$\text{loss}_{\text{flow}} = \int_A \nabla \cdot [f(x)\, p(x,t)]\, \mathrm{d}x.$$

- **Loss due to jumps.** Mass at $x \in A$ jumps away at rate $\lambda(x)$:

$$\text{loss}_{\text{jump}} = \int_A \lambda(x)\, p(x,t)\, \mathrm{d}x.$$

- **Gain due to jumps.** Mass at $y \notin A$ jumps into A with probability $Q(y,A)$ at rate $\lambda(y)$:

$$\text{gain}_{\text{jump}} = \int_E \lambda(y)\, p(y,t)\, Q(y,A)\, \mathrm{d}y = \int_A \int_E \lambda(y)\, p(y,t)\, Q(y,\mathrm{d}z)\, \mathrm{d}x.$$

Kolmogorov Forward Equation: Formal Derivation

Starting from the Dynkin formula, for any $\psi \in \mathcal{D}(\mathcal{A})$:

$$\frac{d}{dt}E[\psi(X_t)] = E[\mathcal{A}\psi(X_t)] = \int_E \mathcal{A}\psi(x)\, p(x,t)\, \mathrm{d}x.$$

But also $\frac{d}{dt}E[\psi(X_t)] = \int_E \psi(x)\, \partial_t p(x,t)\, \mathrm{d}x$. So we need the L^2-adjoint $\mathcal{A}^*$ satisfying $\int \psi\, \mathcal{A}^* p = \int \mathcal{A}\psi \cdot p$ for all ψ.

ODE term. Integration by parts (no boundary terms under Davis's assumption):

$$\int_E f(x) \cdot \nabla\psi(x)\, p(x,t)\, \mathrm{d}x = -\int_E \psi(x)\, \nabla \cdot [f(x)\, p(x,t)]\, \mathrm{d}x.$$

Jump term. Expanding and swapping the role of x and y in the gain part:

$$\int_E \lambda(x) \int_E [\psi(y) - \psi(x)]\, Q(x, \mathrm{d}y)\, p(x,t)\, \mathrm{d}x$$
$$= \int_E \psi(x) \left[\int_E \lambda(y)\, p(y,t)\, Q(y, \mathrm{d}x) - \lambda(x)\, p(x,t) \right] \mathrm{d}x.$$

Kolmogorov Forward Equation: Result

Theorem (Kolmogorov Forward Equation)

Assuming X_t has density $p(x,t)$, it satisfies:

$$\boxed{\partial_t p(x,t) = \underbrace{-\nabla \cdot [f(x)\, p(x,t)]}_{\textit{transport}} \underbrace{- \lambda(x)\, p(x,t)}_{\textit{loss via jumps}} + \underbrace{\int_E \lambda(y)\, p(y,t)\, Q(y, \mathrm{d}x)}_{\textit{gain via jumps}}}$$

Transport term

First-order hyperbolic PDE. No diffusion. Solved by method of characteristics along $\dot{x} = f(x)$.

Loss term

Probability mass leaves x at rate $\lambda(x)$. Local operator.

Gain term

Non-local integral operator. Couples the density at all states y jumping into x.

Stationary Condition

Setting $\partial_t p = 0$ in the Kolmogorov forward equation, a density $\pi(x)$ is stationary if and only if:

$$\boxed{\nabla \cdot [f(x)\,\pi(x)] + \lambda(x)\,\pi(x) = \int_E \lambda(y)\,\pi(y)\,Q(y, \mathrm{d}x)}$$

- The left-hand side is a local operator: transport out of x plus probability mass lost via jumps at x.
- The right-hand side is non-local: probability mass arriving at x from jumps originating at all $y \in E$.
- This is an integro-differential equation — generally hard to solve analytically.

Ergodicity

Definition

A PDMP with semigroup (P_t) is **ergodic** if it admits a unique invariant probability measure π, satisfying

$$\int_E P_t\psi \, \mathrm{d}\pi = \int_E \psi \, \mathrm{d}\pi \quad \text{for all bounded measurable } \psi, \ t \geq 0,$$

and for all initial conditions $x \in E$:

$$\mu_t \xrightarrow{t\to\infty} \pi \quad \text{weakly},$$

where $\mu_t = P_t(x, \cdot)$ is the distribution of X_t starting from $X_0 = x$.

Ergodicity

Stronger notions

- **Positive recurrence:** the process returns to recurrent sets in finite expected time (defined precisely after petite sets are introduced).

- **Geometric ergodicity:** convergence is exponentially fast in a weighted norm $\|\cdot\|_V$ for some $V \geq 1$.

The Ergodic Theorem

Theorem (Ergodic theorem, Meyn & Tweedie (1993) Theorem 13.0.1)

Let the PDMP be ergodic with invariant measure π. Then for all $x \in E$ and all $g \in L^1(\pi)$:

$$\frac{1}{T}\int_0^T g(X_t)\,\mathrm{d}t \xrightarrow{T\to\infty} \int_E g\,\mathrm{d}\pi \quad a.s.$$

No rate without more structure

Ergodicity gives convergence but no rate. Positive recurrence and geometric ergodicity provide quantitative control on how fast $\mu_t \to \pi$.

Irreducibility: A Necessary Condition

Definition (ν-irreducibility, Meyn & Tweedie (1993) Chapter 4)

A PDMP is ν-**irreducible** with respect to a σ-finite measure ν on E if for every $x \in E$ and every measurable A with $\nu(A) > 0$:

$$\int_0^\infty P_t(x, A)\,\mathrm{d}t > 0.$$

What this requires for a PDMP

- **ODE reachability.** From x, the flow $\varphi(t, x)$ must reach a neighborhood of some point y from which a jump into A is possible.
- **Jump support.** $Q(y, A) > 0$ for some y reachable by the flow from x.

Irreducibility: Necessary but Not Sufficient

Irreducibility is necessary but not sufficient

Without irreducibility, ergodicity cannot hold: the process could be confined to a strict subset of E forever. However, irreducibility alone gives no information about convergence — additional conditions are needed.

When irreducibility fails: limit cycles

If the ODE has an invariant set $\mathcal{L} \subset E$ (e.g. a limit cycle) and $Q(x, \mathcal{L}) = 1$ for all $x \in \mathcal{L}$, then the process started on $\mathcal{L}$ can never leave it:

$$P_x(X_t \in \mathcal{L} \text{ for all } t \geq 0) = 1, \quad x \in \mathcal{L}.$$

Any A disjoint from $\mathcal{L}$ satisfies $\int_0^\infty P_t(x, A)\,\mathrm{d}t = 0$, so ν-irreducibility fails.

Harris Recurrence: Returning Infinitely Often

Irreducibility guarantees the process can reach any set, but not that it returns *infinitely often*.

Definition (Harris recurrence, Meyn & Tweedie (1993) Chapter 9)

A ν-irreducible PDMP is **Harris recurrent** if for every measurable A with $\nu(A) > 0$:

$$P_x\left(\int_0^\infty \mathbf{1}_{X_t \in A}\,\mathrm{d}t = \infty\right) = 1 \quad \text{for all } x \in E.$$

Why Harris recurrence is needed

Without it, the process could visit every set at least once but then drift away to infinity, never to return. Harris recurrence rules this out: every positive-measure set is visited infinitely often with probability 1.

Harris Recurrence: Connection to Ergodicity

Harris Ergodic Theorem,Meyn & Tweedie (1993), Theorem 13.0.1.

A ν-irreducible, Harris recurrent, positively recurrent PDMP admits a unique invariant measure π and satisfies

$$\mu_t \xrightarrow{t\to\infty} \pi \quad \text{weakly}, \quad \text{for } \textit{all } x \in E.$$

The convergence holds for *every* starting point $x \in E$, not just π-almost every one. This is the key advantage of Harris recurrence over standard ergodicity: it gives a *uniform* guarantee across all initial conditions.

What is still missing

Positive recurrence has not yet been defined — it requires the notion of a **petite set**, introduced next.

Petite Sets: Definition

Definition (Petite set, Meyn & Tweedie (1993) Chapter 5)

A measurable set $C \subset E$ is **petite** if there exist $T > 0$, $\varepsilon > 0$ and a probability measure ν_C on E such that:

$$P^T(x, \cdot) \geq \varepsilon \, \nu_C(\cdot) \quad \text{for all } x \in C.$$

Intuition: the refreshment mechanism

Every visit to C carries a probability ε of *refreshing* the distribution to ν_C, regardless of history. This breaks dependence on the initial condition and forces convergence to π.

Without petite sets, two trajectories starting at different points could both visit C infinitely often yet never couple — their distributions would never mix.

Petite Sets: Verification for PDMPs

Sufficient conditions for a compact set C to be petite, Costa & Dufour (1999) Section 3

- **Jump rate bounded below:** $\lambda(x) \geq \lambda_{\min} > 0$ for all $x \in C$. Ensures jumps occur at a positive rate throughout C.
- **Jump kernel has a density bounded below:** $Q(x, \cdot)$ has a density bounded away from zero on C. Ensures jumps spread mass uniformly.
- **Local ODE controllability:** the flow connects points in C, so the process can reach any part of C from any other part.

Positive Recurrence

Definition (Positive recurrence, Meyn & Tweedie (1993) Theorem 10.0.1)

A Harris recurrent PDMP is **positively recurrent** if for some petite set C with $\nu(C) > 0$:

$$E_x[\tau_C] < \infty \quad \text{for all } x \in E,$$

where $\tau_C = \inf\{t > 0 : X_t \in C\}$.

Independence of the choice of C

If $E_x[\tau_C] < \infty$ holds for one petite set C, it holds for *all* petite sets with $\nu(C) > 0$. The definition is therefore well-posed.

Positive Recurrence: Connection to π

Invariant measure via excursions, Meyn & Tweedie (1993) Theorem 10.0.1

Positive recurrence is equivalent to the existence of a finite invariant measure π. Moreover:

$$\pi(A) \propto E_x\left[\int_0^{\tau_C} \mathbf{1}_{X_t \in A}\, \mathrm{d}t\right],$$

i.e. $\pi(A)$ is proportional to the expected time spent in A during one excursion from C.

Interpretation

The invariant measure π is built from the local behaviour of the process during a single return cycle to C. This gives a constructive characterisation of π that does not require solving the stationary equation directly.

Summary so far

We have shown that:

$$\nu\text{-irreducibility} + \text{Harris recurrence} + \text{positive recurrence} \implies \text{ergodicity}.$$

The next slides give *sufficient conditions* for all three to hold simultaneously.

A Sufficient Condition for Ergodicity: Foster's Criterion

Theorem (Foster's criterion, Meyn & Tweedie (1993) Theorem 3.2; Costa & Dufour (1999) Theorem 3.1)

Suppose the PDMP is ν-irreducible. If there exist $V \geq 1$, a petite set C, and constants $\varepsilon > 0$, $b < \infty$ such that:

$$\mathcal{A}V(x) \leq -\varepsilon + b\,\mathbf{1}_C(x),$$

then the PDMP is Harris recurrent, positively recurrent, and admits a unique invariant distribution π.

Interpretation: Outside C, the expected infinitesimal change in V is strictly negative — the process is pushed back towards C at a uniform rate ε. Inside C, V may increase by at most b. The Lyapunov function V measures how far the process is from the petite set C.

Foster's Criterion: Bound on Return Times

Derivation via the Dynkin formula

Apply the Dynkin formula to V up to τ_C. Since $C_t^V = V(X_t) - V(X_0) - \int_0^t \mathcal{A}V(X_s)\,\mathrm{d}s$ is a local martingale, taking expectations and using optional stopping at τ_C:

$$E_x[V(X_{\tau_C})] - V(x) = E_x\left[\int_0^{\tau_C} \mathcal{A}V(X_s)\,\mathrm{d}s\right] \leq -\varepsilon\, E_x[\tau_C] + b.$$

Since $V \geq 1$ and $E_x[V(X_{\tau_C})] \geq 1$:

$$\varepsilon\, E_x[\tau_C] \leq V(x) - 1 + b,$$

so:

$$E_x[\tau_C] \leq \frac{V(x) - 1 + b}{\varepsilon} < \infty \quad \text{for all } x \in E.$$

Consequence

A Sufficient Condition for Geometric Ergodicity

Theorem (Meyn & Tweedie (1993), Durmus, Guillin & Monmarché (2021))

Suppose the PDMP is ν-irreducible. If there exist $V \geq 1$, a petite set C, and constants $c > 0$, $b < \infty$ such that:

$$\mathcal{A}V(x) \leq -c\, V(x) + b\, \mathbf{1}_C(x),$$

then the PDMP is geometrically ergodic: there exist $C_0 < \infty$ and $\varrho > 0$ such that

$$\|\mu_t - \pi\|_V \leq C_0\, V(x_0)\, e^{-\varrho t}.$$

Strengthening of Foster's criterion

The $-cV$ drift forces faster returns to C for states with large V: the further the process is from C, the stronger the pull back. This quantitative control on return times delivers exponential convergence.

TCP/IP Congestion Control: The Model

Local characteristics

- **State space:** $E = \mathbb{R}_+$, where x is the current sending rate. No reachable boundary: $t^*(x) = \infty$ for all $x \in E$.
- **Flow:** $f(x) = c > 0$ (additive increase), so $\varphi(t, x) = x + ct$.
- **Jump rate:** $\lambda(x) = \alpha x$, $\alpha > 0$ — loss probability grows with the sending rate.
- **Jump kernel:** $Q(x, \cdot) = \delta_{x/2}$ — upon a loss event the rate is halved.

Generator

For $\psi \in \mathcal{D}(\mathcal{A})$:

$$\mathcal{A}\psi(x) = c\,\psi'(x) + \alpha x \left[\psi\left(\tfrac{x}{2}\right) - \psi(x)\right]$$

TCP/IP: Irreducibility

Claim

The TCP/IP PDMP is ν-irreducible with respect to the Lebesgue measure ν on $\mathbb{R}_+$.

Argument. Fix $x > 0$ and any interval $(a, b) \subset \mathbb{R}_+$.

- **ODE reachability.** The flow $\varphi(t, x) = x + ct$ is strictly increasing, so from x the process can reach any value $y > x$ in finite time $t = (y - x)/c$.
- **After one jump.** From y, a jump sends the process to $y/2$. By choosing $y = 2z$ for any $z \in (a, b)$, the process lands in (a, b) after one jump.
- **For $x > 2b$.** First flow to $y = 2z \in (2a, 2b)$, then jump to $z \in (a, b)$. Since $\lambda(y) = \alpha y > 0$, this occurs with positive probability.

Hence $\int_0^\infty P_t(x, (a, b))\, \mathrm{d}t > 0$ for all $x > 0$ and all (a, b), so ν-irreducibility holds.

TCP/IP: Petite Set

Claim

For any $0 < a < b < \infty$, the compact set $C = [a, b]$ is petite.

Argument. We need to find $T > 0$, $\varepsilon > 0$ and ν_C such that $P^T(x, \cdot) \geq \varepsilon\, \nu_C(\cdot)$ for all $x \in C$.
Since $Q(x, \cdot) = \delta_{x/2}$ is a point mass, we argue via **two jumps**:

- After the first jump from $x \in [a, b]$, the process lands at $x/2 \in [a/2, b/2]$ and then evolves under the flow.
- After a second jump from some $y \in [a/2 + ct, b/2 + ct]$, the process lands at $y/2$, which has an **absolutely continuous** distribution since y varies continuously with the flow time t.
- On C: $\lambda(x) \geq \alpha a > 0$, so both jumps occur at a positive rate.

After two jumps, the distribution of X has an absolutely continuous component bounded below by some $\varepsilon\, \nu_C$ on a compact set. Hence C is petite.

TCP/IP: Ergodicity via Foster's Criterion

Lyapunov function: $V(x) = x$

$$\mathcal{A}V(x) = c \cdot 1 + \alpha x \left[\frac{x}{2} - x\right] = c - \frac{\alpha x^2}{2}$$

Applying Foster's criterion. Choose $\delta > 0$ small and set $C = \left[\delta, \sqrt{\frac{2c}{\alpha}} + \delta\right]$. Then:

- **Outside** C, $x > \sqrt{2c/\alpha} + \delta$:

$$\mathcal{A}V(x) = c - \frac{\alpha x^2}{2} \leq c - \frac{\alpha}{2}\left(\sqrt{\frac{2c}{\alpha}} + \delta\right)^2 = -\alpha\delta\sqrt{\frac{c}{2\alpha}} - \frac{\alpha\delta^2}{2} =: -\varepsilon < 0.$$

- **Inside** C: $\mathcal{A}V(x) \leq c =: b < \infty$.

Hence $\mathcal{A}V(x) \leq -\varepsilon + b\,\mathbf{1}_C(x)$ with $\varepsilon > 0$ and $b = c < \infty$.

TCP/IP: Ergodicity via Foster's Criterion

Conclusion:

Since the PDMP is ν-irreducible and C is a petite set, Foster's criterion (Meyn & Tweedie 1993, Theorem 3.2) gives Harris recurrence, positive recurrence, and existence of a unique stationary distribution π.

TCP/IP: Stationary Forward Equation

The stationary Kolmogorov forward equation for the density $\pi(x)$ is:

$$\boxed{c\,\pi'(x) + \alpha x\,\pi(x) = 4\alpha x\,\pi(2x)}$$

Derivation of the right-hand side. The gain term is:

$$\int_0^\infty \lambda(y)\,\pi(y)\,Q(y,\mathrm{d}x) = \int_0^\infty \alpha y\,\pi(y)\,\delta_{y/2}(\mathrm{d}x)\,\mathrm{d}y.$$

The measure $\delta_{y/2}(\mathrm{d}x)$ concentrates at $y = 2x$, so changing variables $y = 2x$, $\mathrm{d}y = 2\,\mathrm{d}x$:

$$= \alpha \cdot 2x \cdot \pi(2x) \cdot 2 = 4\alpha x\,\pi(2x).$$

Non-locality

The equation couples $\pi(x)$ at scale x to $\pi(2x)$ at scale $2x$ — a **functional differential equation**. Standard ODE methods and method of characteristics do not apply directly. The natural tool is the **Mellin transform** (Baccelli, 2002).

TCP/IP: Second Moment in Steady State

Since the PDMP is ergodic with stationary distribution π, for any $\psi \in \mathcal{D}(\mathcal{A})$:

$$\int_E \mathcal{A}\psi \, \mathrm{d}\pi = 0.$$

Choose $\psi(x) = x$. Recall:

$$\mathcal{A}\psi(x) = c - \frac{\alpha x^2}{2}.$$

Applying $\int \mathcal{A}\psi \, \mathrm{d}\pi = 0$:

$$c - \frac{\alpha}{2} E_\pi[X^2] = 0 \implies \boxed{E_\pi[X^2] = \frac{2c}{\alpha}.}$$

Mathis formula revisited...

Bound on $E_\pi[X]$ via Jensen's inequality

Since $x \mapsto x^2$ is convex, Jensen's inequality gives:

$$E_\pi[X]^2 \leq E_\pi[X^2] = \frac{2c}{\alpha},$$

hence:

$$E_\pi[X] \leq \sqrt{\frac{2c}{\alpha}}.$$

Remark: The exact value of $E_\pi[X]$ requires more advanced techniques (Mellin transform).

Summary

Foundations (Davis 1984)

- PDMPs: deterministic flow punctuated by random jumps, characterised by (φ, λ, Q).
- Extended generator $\mathcal{A}$: ODE drift + non-local jump term.
- Kolmogorov forward equation: hyperbolic PDE + integral operator.
- Dynkin formula: bridge between $\mathcal{A}$ and martingales.

Ergodicity

- Irreducibility + Harris recurrence + positive recurrence $\implies$ ergodicity.
- Foster's criterion: sufficient condition via $\mathcal{A}V \leq -\varepsilon + b\mathbf{1}_C$.
- Lyapunov criterion: geometric ergodicity via $\mathcal{A}V \leq -cV + b\mathbf{1}_C$.

TCP/IP example

- Ergodicity verified via Foster's criterion.
- $E_\pi[X^2] = 2c/\alpha$ from the generator.
- $E_\pi[X] \leq \sqrt{2c/\alpha}$ via Jensen.

References I

- M. H. A. Davis. *Piecewise-deterministic Markov processes: A general class of non-diffusion stochastic models.* J. R. Statist. Soc. B **46**(3):353–388, 1984.
- M. H. A. Davis. *Markov Models and Optimization.* Chapman & Hall, 1993.
- O. L. V. Costa. *Stationary distributions for piecewise-deterministic Markov processes.* J. Appl. Prob. **27**:60–73, 1990.
- R. L. Tweedie. *Sufficient conditions for ergodicity and recurrence of Markov chains on a general state space.* Stoch. Proc. Appl. **3**:385–403, 1975.
- D. Vermes. *Optimal control of piecewise deterministic processes.* Stochastics **14**:165–208, 1985.
- F. Dufour, O. L. V. Costa. *Stability of piecewise deterministic Markov processes.* SIAM J. Control Optim. **37**(5):1483–1502, 1999.
- A. Durmus, A. Guillin, P. Monmarche. *Piecewise deterministic Markov processes and their invariant measures.* Ann. Inst. H. Poincaré **57**(3), 2021.

References II

- S. P. Meyn, R. L. Tweedie. *Markov Chains and Stochastic Stability.* Springer-Verlag, London, 1993. (2nd edition: Cambridge University Press, 2009.)
- S. P. Meyn, R. L. Tweedie. *Stability of Markovian processes III: Foster–Lyapunov criteria for continuous-time processes.* Adv. Appl. Prob. **25**(3):518–548, 1993.
- F. Baccelli, D. McDonald, J. Reynier. *A mean-field model for multiple TCP connections through a buffer implementing RED.* Performance Evaluation **49**(1–4):77–97, 2002.
- F. Baccelli, D. R. McDonald. *A stochastic model for the throughput of non-persistent TCP flows.* In *Proceedings of the 1st International Conference on Performance Evaluation Methodologies and Tools* (Valuetools'06), Pisa, Italy, October 2006. ACM, ISBN 1-59593-504-5.

The Stationary Balance Equation

Recall the TCP/IP PDMP: flow $f(x) = c$, jump rate $\lambda(x) = \alpha x$, kernel $Q(x, \cdot) = \delta_{x/2}$.

The stationary Kolmogorov forward equation reduces to:

$$\boxed{c\,\pi'(x) + \alpha x\,\pi(x) = 4\alpha x\,\pi(2x)}$$

The difficulty: scale coupling

The equation couples π at two different scales: $\pi(x)$ on the left and $\pi(2x)$ on the right. This is a **functional differential equation** — standard ODE methods do not apply.

Strategy

The Mellin transform converts the scaling $x \mapsto 2x$ into an algebraic operation, turning the equation into a tractable recurrence.

The Mellin Transform

Definition (Mellin transform)

For $\pi : \mathbb{R}_+ \to \mathbb{R}_+$, the **Mellin transform** is:

$$\hat{\pi}(u) = \int_0^\infty x^{u-1}\pi(x)\,\mathrm{d}x, \quad u \in \mathbb{C},\ \alpha < \mathrm{Re}(u) < \beta.$$

Remark

Since $\hat{\pi}(u) = E_\pi[X^{u-1}]$, integer values give moments directly: $\hat{\pi}(1) = 1$, $\hat{\pi}(2) = E_\pi[X]$, $\hat{\pi}(3) = E_\pi[X^2], \ldots$

The Mellin Transform: Key Properties

1. **Derivative:** $\mathcal{M}[\pi'](u) = -(u-1)\hat{\pi}(u-1)$
   Proof: integration by parts; boundary terms vanish.

2. **Multiplication by** x**:** $\mathcal{M}[x\pi(x)](u) = \hat{\pi}(u+1)$

3. **Scaling:** $\mathcal{M}[\pi(ax)](u) = a^{-u}\hat{\pi}(u) \quad \text{for } a > 0$
   Proof: substitute $y = ax$.

Functional Equation for $\hat{\pi}$

Apply the Mellin transform to both sides of $c\,\pi'(x) + \alpha x\,\pi(x) = 4\alpha x\,\pi(2x)$:

$$\underbrace{-c(u-1)\hat{\pi}(u-1)}_{\text{prop. 1}} + \underbrace{\alpha\hat{\pi}(u+1)}_{\text{prop. 2}} = \underbrace{2^{1-u}\alpha\,\hat{\pi}(u+1)}_{\text{props. 2\&3}}$$

Collecting terms:

$$-c(u-1)\hat{\pi}(u-1) + \alpha(1-2^{1-u})\hat{\pi}(u+1) = 0$$

Functional equation

$$\boxed{\hat{\pi}(u+1) = \frac{c(u-1)}{\alpha(1-2^{1-u})}\,\hat{\pi}(u-1)}$$

A recurrence in steps of 2, valid for all u in the fundamental strip.

The Key Ansatz (Baccelli & McDonald 2006)

Ansatz

Write:

$$\hat{\pi}(u) = g(u)\,\Gamma\left(\frac{u}{2}\right)\left(\frac{2c}{\alpha}\right)^{u/2}$$

Substituting into the functional equation, the Γ and power factors cancel, leaving a simple recurrence for g:

$$\boxed{g(u) = g(u+2)\left(1 - 2^{-u}\right)}$$

Assuming $g(u) \to g(\infty)$ as $\mathrm{Re}(u) \to \infty$ (proved in Baccelli et al. 2005), iterate forward:

$$g(u) = g(\infty)\prod_{k=0}^{\infty}\left(1 - 2^{-u-2k}\right) =: g(\infty)\,\Pi_\infty(u)$$

The Solution for $\hat{\pi}$

Substituting $g(u) = g(\infty)\,\Pi_\infty(u)$ back into the ansatz:

$$\hat{\pi}(u) = C\,\Gamma\left(\frac{u}{2}\right)\left(\frac{2c}{\alpha}\right)^{u/2}\Pi_\infty(u)$$

where $C = g(\infty)$ and:

$$\Pi_\infty(u) = \prod_{k=0}^{\infty}\left(1 - 2^{-u-2k}\right)$$

The Solution for $\hat{\pi}$

Substituting $g(u) = g(\infty)\,\Pi_\infty(u)$ back into the ansatz:

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$$\Pi_\infty(u) = \prod_{k=0}^{\infty}\left(1 - 2^{-u-2k}\right)$$

Verification

$$\frac{\hat{\pi}(u+1)}{\hat{\pi}(u-1)} = \frac{2c}{\alpha}\cdot\underbrace{\frac{\Gamma\left(\frac{u+1}{2}\right)}{\Gamma\left(\frac{u-1}{2}\right)}}_{(u-1)/2}\cdot\underbrace{\frac{\Pi_\infty(u+1)}{\Pi_\infty(u-1)}}_{1/(1-2^{1-u})} = \frac{c(u-1)}{\alpha(1-2^{1-u})}\checkmark$$

where $\Pi_\infty(u-1) = (1-2^{1-u})\Pi_\infty(u+1)$ follows from extracting the $k=0$ term.

Normalization

The constant C is determined by $\hat{\pi}(1) = 1$:

$$1 = C\,\Gamma\left(\frac{1}{2}\right)\left(\frac{2c}{\alpha}\right)^{1/2}\Pi_\infty(1)$$

Using $\Gamma(1/2) = \sqrt{\pi}$:

$$\boxed{C = \frac{1}{\sqrt{\frac{2\pi c}{\alpha}} \cdot \Pi_\infty(1)}}$$

So the normalized Mellin transform is:

$$\hat{\pi}(u) = \frac{\Gamma\left(\frac{u}{2}\right)\left(\frac{2c}{\alpha}\right)^{u/2}\Pi_\infty(u)}{\sqrt{\frac{2\pi c}{\alpha}} \cdot \Pi_\infty(1)}$$

Second Moment: $E_\pi[X^2] = \hat{\pi}(3)$

$$\hat{\pi}(3) = \frac{\Gamma\left(\frac{3}{2}\right)\left(\frac{2c}{\alpha}\right)^{3/2}\Pi_\infty(3)}{\sqrt{\frac{2\pi c}{\alpha}}\cdot\Pi_\infty(1)}$$

Using $\Gamma(3/2) = \sqrt{\pi}/2$ and extracting the $k = 0$ term of $\Pi_\infty(1)$:

$$\Pi_\infty(1) = (1 - 2^{-1})\,\Pi_\infty(3) \implies \frac{\Pi_\infty(3)}{\Pi_\infty(1)} = 2$$

Therefore:

$$\hat{\pi}(3) = \frac{\frac{\sqrt{\pi}}{2}\cdot\left(\frac{2c}{\alpha}\right)^{3/2}\cdot 2}{\sqrt{\frac{2\pi c}{\alpha}}} = \frac{2c}{\alpha}$$

Consistency check $E_\pi[X^2] = 2c/\alpha$, consistent with our earlier generator argument $\int \mathcal{A}\psi \,\mathrm{d}\pi = 0$ with $\psi(x) = x$. ✓

Mean Throughput: $E_\pi[X] = \hat{\pi}(2)$

$$\hat{\pi}(2) = \frac{\Gamma(1)\left(\frac{2c}{\alpha}\right)\Pi_\infty(2)}{\sqrt{\frac{2\pi c}{\alpha}} \cdot \Pi_\infty(1)} = \sqrt{\frac{2c}{\alpha\pi}} \cdot \frac{\Pi_\infty(2)}{\Pi_\infty(1)}$$

Numerically:

$$\frac{\Pi_\infty(2)}{\Pi_\infty(1)} = \frac{\prod_{k=0}^{\infty}(1-4^{-k-1})}{\prod_{k=0}^{\infty}(1-2^{-1-2k})} \approx \frac{0.6894}{0.4278} \approx 1.612$$

Mean throughput in steady state

$$\boxed{E_\pi[X] = \sqrt{\frac{2c}{\alpha\pi}} \cdot \frac{\Pi_\infty(2)}{\Pi_\infty(1)} \approx 0.91\sqrt{\frac{2c}{\alpha}}}$$

Connection to the Mathis Formula

Mathis formula

For a persistent TCP flow with packet loss probability p and round trip time R:

$$M = \frac{K}{R\sqrt{p}}$$

For **periodic losses**, $K = \sqrt{3/2} \approx 1.22$ is derived from the deterministic sawtooth window analysis.

In our model $c = 1/R^2$ and $\alpha = p$, so our formula for **random losses** gives:

$$E_\pi[X] = \frac{1}{R\sqrt{p}} \cdot \sqrt{\frac{2}{\pi}} \cdot \frac{\Pi_\infty(2)}{\Pi_\infty(1)} \approx \frac{1.286}{R\sqrt{p}}$$

which has the same $1/(R\sqrt{p})$ structure as the Mathis formula, with the constant determined by the stochastic loss process.